

Communications System

Software Requirements Specification

Revision History

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1. **Purpose**

This document outlines the requirements for the Communications System. It will explain the purpose and the features of the system. The constraints under which it must operate and how the system is intended to behave. This document is intended for the developers and stakeholders of the system.

1.1. Scope

This document will catalog the user, system, and hardware requirements for the Communications system. It will not, however, document how these requirements will be implemented.

1.2. Definitions, Acronyms, Abbreviations

1.2.1 User Login Module - ULM

1.2.2 Client Module - CM

1.2.3 Communication Server Module - CSM

1.3. References

Use Case Specification Document – Design Specification Document

UML Use Case Diagram Document – Design Specification Document

Class Diagram – Design Specification Document

Sequence Diagrams – Design Specification Document

1.4. Overview

Scalable, text-only communications system designed to sustain sending messages between multiple users. These messages can be sent synchronously and asynchronously in group chats or private chats. Users will be limited to messages that are only for them, and all messages will be viewable by IT staff. Users and IT staff interact with the system through the provided GUI.

2. Overall Description

2.1. Product Perspective

This product is a real-time communication system that runs through a two-way client-server connection. The product allows authenticated users to reach the server and send messages through TCP/IP networking connections. Ultimately, the product should support a large number of concurrent users, with message history stored in a data handling file. The system does not rely on external databases, making it less optimized for retaining large amounts of historical data.

2.2. Product Architecture

The system will be organized into 3 major modules: User Login Module, Client Module, and the Communication Server Module.

The system will be designed so that the ULM is responsible for determining user permissions in the CM. Once that has been determined, the CM will be directly connected to CSM, granting access to messaging services, along with the respective User and IT, and permissions. The CSM provides the main functionality behind the product's design.

The system architecture will be modeled using UML diagrams

2.3. Product Functionality/Features

The high-level features of the system are as follows (see section 3 of this document for more detailed requirements that address these features):

1. User account authentication
2. Real-time text messaging (Synchronous & Asynchronous)
3. One-on-one and group communication
4. IT message log viewing & exporting

2.4. Constraints

1. Application must be developed based on Java
2. The system will not be using databases, libraries, frameworks, or other technologies.
3. The system follows standard object-oriented design principles

2.5. Assumptions and Dependencies

List appropriate assumptions

1. It is assumed that there is no maximum number of users at a given time
2. The hardware running the system can support it

3. Functional Requirements

3.1. Common Requirements:

- 3.1.1. Each User account must have a unique ID, Username, and password which can be used to log in. *Stretch goal:* Username and password are alphanumeric strings between 6 and 20 characters in length.
- 3.1.2. All communications between client and server will use TCP/IP

3.2. User Login Module Requirements:

- 3.2.1. The User Login Module (ULM) contains a login menu to allow users to log into the communications system. This menu is displayed upon loading the client
- 3.2.2. The User Login Module (ULM) contains a button that allows the user to sign up for an account
- 3.2.3. The ULM contains a check box allowing the user to sign up as an IT user
- 3.2.4. Users can log in to their accounts through the login menu by entering a valid username and password.
- 3.2.5. After a user presses login, the client will send the entered username and password to the CSM to be authenticated
- 3.2.6. During the authorization process, the user's role and permissions are determined.
- 3.2.7. On a successful login, the CSM will provide the user with access to their respective role's user interface
- 3.2.8. On a failed login, the CSM communicates the failure to the user, and the ULM prompts the user to re-enter their username and password

3.3. Client Module Requirements:

- 3.3.1. The Client Module (CM) contains a communications menu to allow users to communicate with other users. This menu is displayed upon logging in to their account.
- 3.3.2. The CM must be able to secure a connection to send and receive signals from the CSM
- 3.3.3. The communications menu contains a list of chats the user is a part of
- 3.3.4. The user can select any chat in the chat list to open it
- 3.3.5. The list is ordered from the one with the most recent date contained in the most recent message or creation date, to the one that is the oldest.
- 3.3.6. The server can signal to the client if this list needs to be updated, and in turn, the client will update it, displaying the most up-to-date information
- 3.3.7. The communications menu contains a new chat button, which creates a chat between you and the users specified
- 3.3.8. Only inputting 1 username defaults the chat's designation as a private message chat. A user can select for the chat to be a group chat, designating it as a group message chat
- 3.3.9. Inputting multiple usernames defaults the chat's designation as a group message chat. This cannot be designated as anything other than a group message chat
- 3.3.10. Upon creating the new chat, the client will send the request to the server, creating the chat.
- 3.3.11. The communications menu contains a profile button, displaying the user's username, audit mode, and a button to log out of the account
- 3.3.12. The logout button redirects to the login menu upon being selected
- 3.3.13. The communications menu contains a chat menu, displaying a specific chat, selected from the list of chats the user is involved in
- 3.3.14. The chat selected must be designated as a private or a group chat.
- 3.3.15. In this menu, the user can see a scrollable view of all messages previously sent in the chat.

- 3.3.16. Each message displays the username of the sender, the time it was sent, and the message sent
- 3.3.17. *Optional, if time constraints permit:* Chats containing messages sent while the user was offline are indicated as unread
- 3.3.18. *Optional, if time constraints permit:* When the user's client displays an unread chat, it is no longer indicated as unread
- 3.3.19. Whenever a new message is sent from the server, this view is updated, displaying the new message at the forefront
- 3.3.20. The chat menu contains a message box allowing the user to write a message, and a button to send it.
- 3.3.21. When a message is sent, it is sent to the server to update the clients of the users involved in the chat
- 3.3.22. When a message is sent, the date and time stamp at that moment is sent to the server to ensure messages are displayed in order
- 3.3.23. *Optional, if time constraints permit:* The chat menu displays all users involved with the chat, also indicating if this is a group chat or a private chat
- 3.3.24. The group chat creator's view of the group chat displays an invite, kick, and delete group chat button
- 3.3.25. After pressing the invite button, the group chat creator can enter usernames here to invite users to the chat.
- 3.3.26. After pressing the kick button, the group chat creator is prompted to select the user to kick
- 3.3.27. *Optional, if time constraints permit:* After pressing the delete group chat button, the group chat creator is prompted to confirm the action, and the group chat is deleted
- 3.3.28. A deleted group chat still has its messages logged, but must not be displayed to the related users
- 3.3.29. The IT users will have access to the same permissions as regular users.
- 3.3.30. IT users will have additional features accessible. Access to these permissions is determined in the ULM.
- 3.3.31. The IT users will be provided an audit menu that allows them to review other users' conversations in read-only mode.
- 3.3.32. When the IT opts to change their view into Audit Mode, the application displays a User List screen.
- 3.3.33. *Optional, if time constraints permit:* Through the user list screen, the IT user is able to see a display of a list of available users retrieved from the server.
- 3.3.34. The user list screen includes a search bar that allows the IT user to search for users by username
- 3.3.35. When the IT user selects a user, the application enters Audit Mode for that selected user and displays the selected user's Chat List
- 3.3.36. In Audit Mode, when a chat is opened, the IT user can view the full message history for that chat in a scrollable view.
- 3.3.37. The IT user can view messages from any chats.
- 3.3.38. In Audit Mode, the IT user is not permitted to send, delete, or edit any chats or messages.
- 3.3.39. In Audit Mode, the application visually indicates that the IT user is viewing chats as another user
- 3.3.40. In Audit Mode, there is a button to export the displayed chat. This saves the chat log locally to the IT user as a file
- 3.3.41. The IT user can exit Audit Mode by pressing the exit audit mode button

3.4. Communications Server Module Requirements:

- 3.4.1. The Communications Server Module (CSM) must store messages locally in text files
- 3.4.2. The format of these files will be comma-separated with the following fields: User ID, Username, Message, date, and time stamp.
- 3.4.3. The CSM must be able to secure a connection to the ULM and the CM
- 3.4.4. The CSM must be able to concurrently support multiple ULMs and CMs
- 3.4.5. The CSM stores usernames and passwords for account authentications, to be compared with usernames and passwords sent from the ULM
- 3.4.6. Upon succeeding or failing authentication, the CSM will broadcast to the client the ULM's response.
- 3.4.7. *Optional, if time constraints permit:* Upon succeeding authentications, the CSM marks the user as online
- 3.4.8. The CSM must send the messages to be displayed by the CM in the chronological order of their date and time stamps.
- 3.4.9. The CSM must send to the CM the most up-to-date list of chats for users. This will be updated if the CSM receives a message that would change that status.
- 3.4.10. The CSM must store all messages
- 3.4.11. *Optional stretch goal, if time constraints permit:* The CSM must keep track of messages that were sent to a user while they were offline, to ensure the user is displayed unseen messages in the CM
- 3.4.12. The CSM can send the message logs on request by the CM to authorized IT users.
- 3.4.13. *Optional stretch goal, if time constraints permit:* Upon a user logging out, the CM will signal to the CSM, in which the CSM marks the user as offline.

3.5. External Interface Requirements

- 3.5.1. This section specifies the external interfaces between the Communications System and external entities, including client devices, network protocols, and persistent storage files
- 3.5.2. Client Application User Interface: The system shall provide a graphical user interface for the CM and ULM to support login, chat selection, message viewing, and message sending.
- 3.5.3. The client interface shall display a login menu that accepts username and password input and displays login success or failure feedback.
- 3.5.4. The client interface shall provide a communications menu that displays a list of chats the user is a member of, including a specific indication of unread messages.
- 3.5.5. The client interface shall provide a chat view that displays message history in the selected chat, including sender username, timestamp, and message content.
- 3.5.6. The client interface shall provide a message input box and a send action for transmitting messages to the server.
- 3.5.7. The client interface shall provide a logout function that ends the session and returns the user to the login menu.
- 3.5.8. The client interface shall logout if automatically if the user closes the window that ends the session.
- 3.5.9. Network Communication Interface: All communication between the client and the communication service module shall use TCP/IP.
- 3.5.10. The system shall support a request/response messaging interface for chat creation, message sending, and message retrieval.
- 3.5.11. *Optional stretch goal, if time constraints permit:* The system shall support server-to-client notifications for events, including new message received and unread message indicators.
- 3.5.12. The system shall detect and handle client disconnections and reconnections.

- 3.5.13. Storage Interface (File-Based): The CSM shall store message history in a persistent document/file format (no database).
- 3.5.14. The system shall write stored messages in a format that preserves message ordering within each conversation.
- 3.5.15. The storage format shall include the following fields per message: sender username, timestamp, and message content.
- 3.5.16. *Optional stretch goal, if time constraints permit:* The system shall support offline message storage such that messages sent while a user is offline are available upon the user's next login.

3.6. Internal Interface Requirements

- 3.6.1. The CSM shall provide internal service interfaces for the ULM and CM to request login authentication, messaging, and IT Auditing.
- 3.6.2. The ULM shall send authentication requests containing username and password to the communication server module.
- 3.6.3. The CSM shall validate authentication requests and return authentication results.
- 3.6.4. With successful authentication, the CSM shall return the authenticated user role and session information to the requesting module.
- 3.6.5. The CSM shall provide CM service to create chats, add users to chats, and retrieve user chat lists.
- 3.6.6. The CSM shall provide a method to submit a message to a chat.
- 3.6.7. The CSM shall provide a method to retrieve message history, retrieve unread messages, and mark messages as read.
- 3.6.8. The CSM shall notify the CM when new messages are received
- 3.6.9. The CSM shall provide internal access to IT users and provide access to the chat list of all users, and the contents of each chat within those lists.
- 3.6.10. The CSM shall handle concurrent service requests from multiple ULMs and CMs.
- 3.6.11. The CSM shall provide authorized IT users with access to the data required for Audit Mode.
- 3.6.12. Communication between the CSM, ULM, and CM modules will be done through a message object
- 3.6.13. The message object contains the type of communication and the necessary data for each type of message
- 3.6.14. The type of the message is split up into the main type and the subtype.
- 3.6.15. The CSM and the Client Modules each have a Connection Handler, which determines what message should be sent, and the logic of what to do on receiving a message
- 3.6.16. All internal service interfaces between modules shall follow standard object-oriented interface design.

4. Non-Functional Requirements

4.1. Security and Privacy Requirements

- 4.1.1. Authentication must be done before accessing any communication function.
- 4.1.2. The system shall prevent unauthorized access to chats.
- 4.1.5. All communication between client and server shall be via TCP/IP.
- 4.1.6. Users shall only view conversations in which they are participants.
- 4.1.7. IT users shall be able to view all conversations.
- 4.1.8. The order of text messages shall not be modified.
- 4.1.9. Text messages sent cannot be modified.

4.2. Environmental Requirements

- 4.2.1. The system shall be developed and implemented using Java.
- 4.2.2. Any database systems shall be prohibited.
- 4.2.3. The system shall not use any encryption technology.
- 4.2.4. The system shall be able to operate on a standard operating system (Windows, Linux, MacOS)
- 4.2.5. The system shall operate in TCP/IP network environments.

4.3. Performance Requirements

- 4.3.1. Login response time shall be within 3-5 seconds.
- 4.3.2. Text message delivery time shall be within 3-5 seconds.
- 4.3.3. Chat list updates shall be within 3-5 seconds.
- 4.3.4. The system shall be able to support as many chats and messages as the hardware allows
- 4.3.5. The system shall reserve messages even if the server restarts.
- 4.3.7. The interface shall be intuitive and consistent.